Terraform – 6

Create a new folder on your Desktop: terraform-06

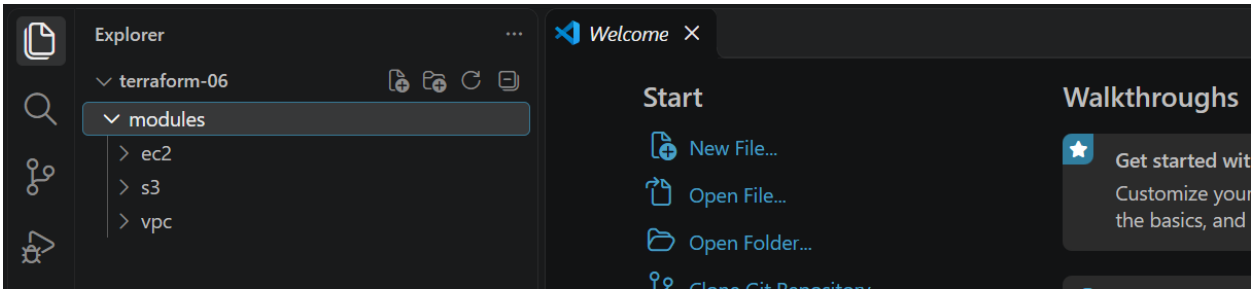
mkdir modules

mkdir modules\ec2

mkdir modules\s3

mkdir modules\vpc

run these PowerShell commands:



Open `modules/ec2/main.tf` and paste:

```
resource "aws_instance" "web" {
  ami           = var.ami_id
  instance_type = var.instance_type

  tags = {
    Name = var.instance_name
  }
}
```

Now open `modules/ec2/variables.tf` and paste:

```
variable "ami_id" {
  description = "AMI ID for EC2 instance"
  type        = string
```

```
}

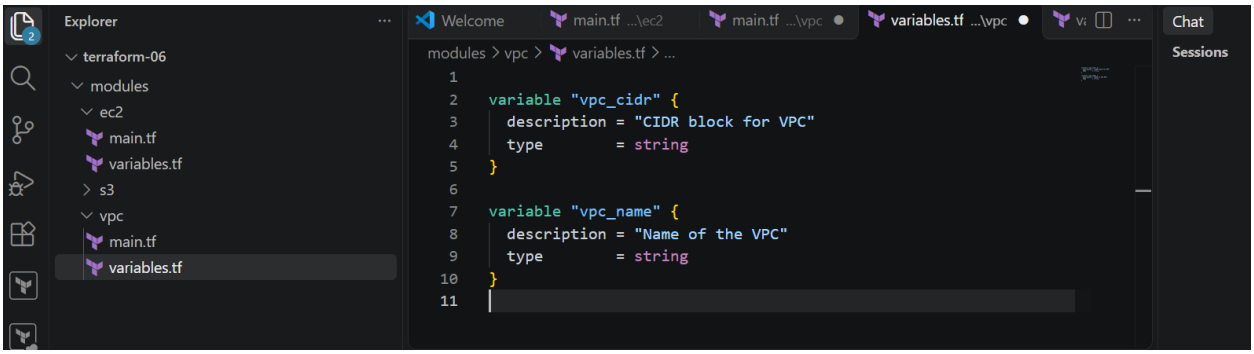

variable "instance_type" {
  description = "EC2 instance type"
  type        = string
}

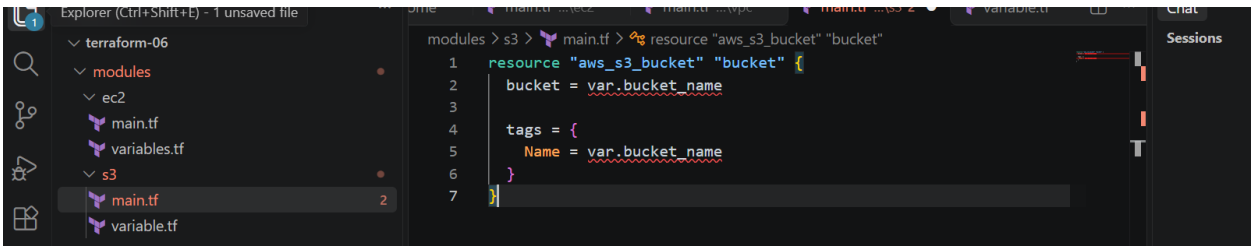

variable "instance_name" {
  description = "EC2 instance name"
  type        = string
}
```

Open the VPC `main.tf` and paste:

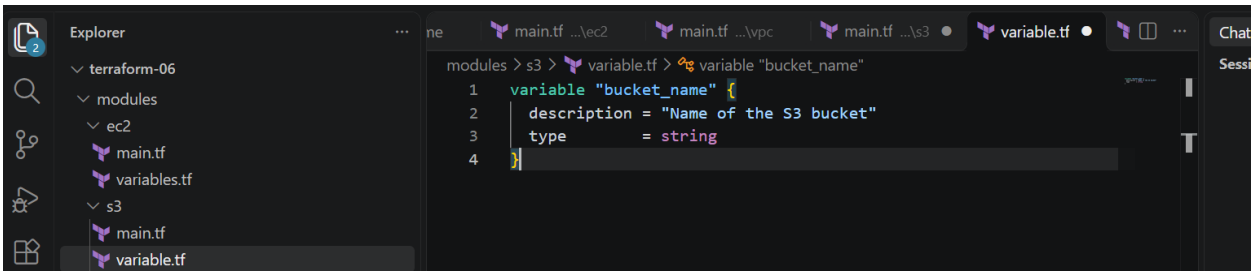


modules/vpc/variables.tf





Add in the s3 main



Add in the s3 variabe

Your three reusable modules are now created:

EC2 module
VPC module
S3 module

Now we need to connect them through the root module.

Prepare the VPC module for EC2

Replave vpc/main.tf with

```
resource "aws_vpc" "main" {
 cidr_block = var.vpc_cidr


 tags = {
  Name = var.vpc_name
 }
}


resource "aws_subnet" "public" {
 vpc_id              = aws_vpc.main.id
 cidr_block          = var.subnet_cidr
 availability_zone   = var.availability_zone
 map_public_ip_on_launch = true


 tags = {
  Name = "${var.vpc_name}-public-subnet"
 }
```

```
}

resource "aws_internet_gateway" "igw" {
  vpc_id = aws_vpc.main.id

  tags = {
    Name = "${var.vpc_name}-igw"
  }
}

resource "aws_route_table" "public" {
  vpc_id = aws_vpc.main.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.igw.id
  }
}

resource "aws_route_table_association" "public" {
  subnet_id      = aws_subnet.public.id
  route_table_id = aws_route_table.public.id
}
```

And variable with

```
variable "vpc_cidr" {
  type = string
}
```

```
variable "vpc_name" {
  type = string
}

variable "subnet_cidr" {
  type = string
}

variable "availability_zone" {
  type = string
}
```

This will create a VPC, public subnet, Internet Gateway and public route table.

VPC module now includes:

VPC

Public subnet

Internet Gateway

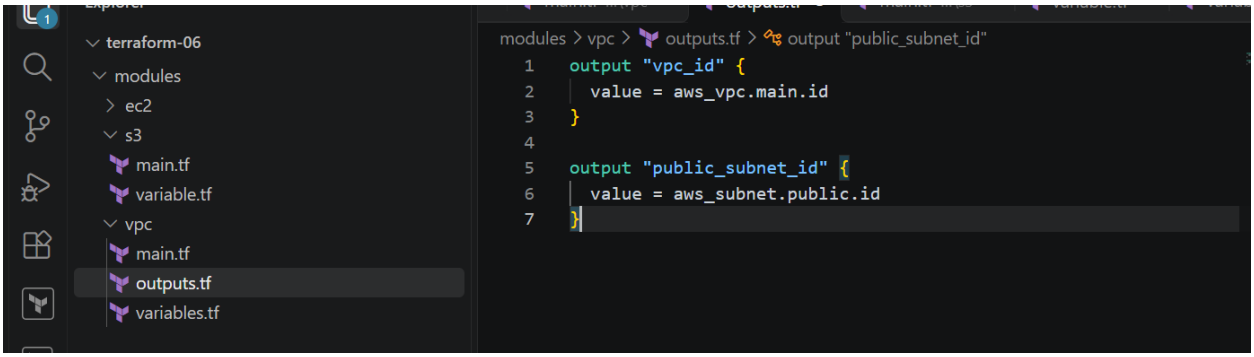
Public route table

Route table association

Now we need to connect the VPC module to the EC2 module.

Create VPC outputs

Inside `modules/vpc`, create a new file named `outputs.tf`.



Why do we need outputs?

The VPC module creates the VPC and subnet. Outputs allow the root module to retrieve their IDs and pass them to the EC2 module.

VPC module now exposes the VPC ID and public subnet ID.

Now create main.tf file in terraform folder, it should look like

terraform-06/

|

├── main.tf ← Root module

|

└── modules/

├── ec2/

| ├── main.tf

| └── variables.tf

├── s3/

| ├── main.tf

| └── variable.tf

└── vpc/

├── main.tf

├── variables.tf

└── outputs.tf

Copy the script

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 6.0"
    }
  }
}

provider "aws" {
  region = "us-east-1"
}

module "vpc" {
  source = "./modules/vpc"

  vpc_cidr         = "10.20.0.0/16"
  vpc_name         = "Terraform-06-VPC"
  subnet_cidr      = "10.20.1.0/24"
  availability_zone = "us-east-1a"
}

module "ec2" {
```

```
  source = "./modules/ec2"


  ami_id        = "ami-0c230065b91270bc6"
  instance_type = "t3.micro"
  instance_name = "Terraform-06-EC2"
  subnet_id     = module.vpc.public_subnet_id
}


module "s3" {
  source = "./modules/s3"


  bucket_name = "jagan-terraform06-modules-0306"
}
```

```
module.vpc.aws_vpc.main: Creating...
module.s3.aws_s3_bucket.bucket: Creating...
module.vpc.aws_vpc.main: Creation complete after 4s [id=vpc-00dee9f733467f15c]
module.vpc.aws_internet_gateway.igw: Creating...
module.vpc.aws_subnet.public: Creating...
module.vpc.aws_internet_gateway.igw: Creation complete after 2s [id=igw-039d2384bc6e9b77d]
module.vpc.aws_route_table.public: Creating...
module.s3.aws_s3_bucket.bucket: Creation complete after 7s [id=jagan-terraform06-modules-0306]
module.vpc.aws_route_table.public: Creation complete after 3s [id=rtb-06f1ac587deb72e29]
module.vpc.aws_subnet.public: Still creating... [00m10s elapsed]
module.vpc.aws_subnet.public: Creation complete after 13s [id=subnet-07f5e956af071030f]
module.vpc.aws_route_table_association.public: Creating...
module.ec2.aws_instance.web: Creating...
module.vpc.aws_route_table_association.public: Creation complete after 1s [id=rtbassoc-05506052bd7a26d67]
module.ec2.aws_instance.web: Still creating... [00m10s elapsed]
module.ec2.aws_instance.web: Creation complete after 15s [id=i-09698a1887de28c5a]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.
```

Created reusable Terraform modules.

Connected modules using `module` blocks.

Passed variables from root module to child modules.

Provisioned VPC, EC2 and S3 using Terraform.

<u>Terraform Workspaces:</u>

create three environments:

1. **Development**
2. **Staging**
3. **Production**

First, let's verify your current workspace.

terraform workspace list

```
Apply complete! Resources: 7 added, 0 changed, 0 destroyed.
PS C:\Users\india\OneDrive\Desktop\terraform-06> terraform workspace list
* default

PS C:\Users\india\OneDrive\Desktop\terraform-06>
```

```
PS C:\Users\india\OneDrive\Desktop\terraform-06> terraform workspace new dev
Created and switched to workspace "dev"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Users\india\OneDrive\Desktop\terraform-06>
```

```
PS C:\Users\india\OneDrive\Desktop\terraform-06> terraform workspace new staging
Created and switched to workspace "staging"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Users\india\OneDrive\Desktop\terraform-06>
```

```
PS C:\Users\india\OneDrive\Desktop\terraform-06> terraform workspace new prod
Created and switched to workspace "prod"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Users\india\OneDrive\Desktop\terraform-06>
```

```
PS C:\Users\india\OneDrive\Desktop\terraform-06> terraform workspace list
  default
  dev
* prod
  staging

PS C:\Users\india\OneDrive\Desktop\terraform-06>
```

Create `main.tf` inside `workspace-ec2`

In VS Code:

1. **Right-click the `workspace-ec2` folder.**
2. **Select New File.**
3. **Name it `main.tf`.**

4. **Paste this code:**

```
data "aws_vpc" "existing" {
  filter {
    name   = "tag:Name"
    values = ["Terraform-06-VPC"]
  }
}

data "aws_subnet" "existing" {
  filter {
    name   = "tag:Name"
    values = ["Terraform-06-VPC-public-subnet"]
  }

  vpc_id = data.aws_vpc.existing.id
}

module "ec2" {
  source = "../modules/ec2"

  ami_id        = "ami-0c230065b91270bc6"
  instance_type = "t3.micro"
  instance_name = "Terraform-06-${terraform.workspace}"
  subnet_id     = data.aws_subnet.existing.id
}
```

```
Initializing the backend...

Initializing modules...
- ec2 in ..\modules\ec2

Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 6.0"...
- Installing hashicorp/aws v6.71.0...
- Installed hashicorp/aws v6.71.0 (signed by HashiCorp)

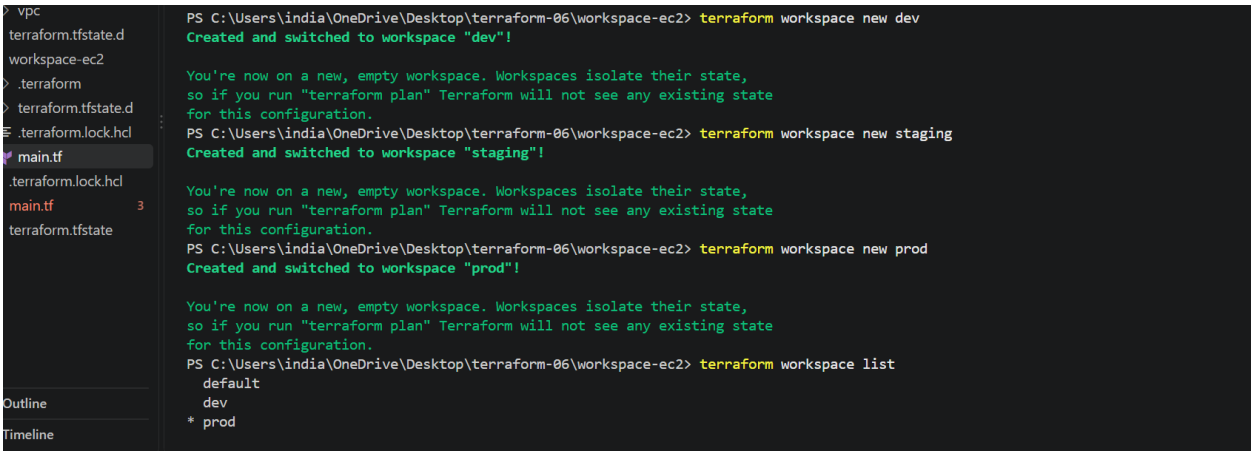
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
```

Now we're working inside `workspace-ec2`.

Important: This folder has its own Terraform workspace list, separate from the parent `terraform-06` folder.



Dev workspace plan is successful.

1 EC2 instance to create

I checked the output:

- **VPC: Existing VPC found successfully.**
- **Subnet: Existing public subnet found successfully.**
- **Instance type: `t3.micro`**
- **Instance name: `Terraform-06-dev`**
- **Planned changes: 1 to add, 0 to change, 0 to destroy.**

Create the Dev EC2 instance

Use terraform apply

